



ŘEDITELSTVÍ SILNIC A DÁLNIC ČR

**NDIC - DATEX II Elaborated Data
Publication - FCD**

Release 1.0rev2

National Traffic Information Centre

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This specification defines format, used for publishing of FCD generated data, e.g. travel times, speed, traffic statuses and quality parameters on a set of predefined locations.

INTRODUCTION

This document shall be used by publisher and / or subscriber. Subscriber shall use it for format suitability evaluation and for possible implementation of received NDIC data import into own information system. Publisher shall use it as a specification of data to be published format and for possible implementation of data export from own information system into NDIC.

Here you can find:

- conceptual description of published information
- information about location referencing methods used
- W3C XML schema of the format
- data sample(s)

Following topics are out of scope of this document and if present, shall be considered as informative only:

- transfer protocol incl. access point, life cycle etc.
- related data formats
- typical data origin
- typical data use

1.1 General terms

Following terms are used in this document:

format

description of data structure for transferring some kind of information

protocol

description of data transfer process and rules

NDIC

national traffic information centre

DATEX II model

class model for describing information related to the road traffic. The model in form of UML is platform independent.

One of specific platforms for expressing the model is XML.

The model is extensive (it consists of few thousands of structural items representing term incl. definition) For reference data (UML class model in form of Enterprise Architect EAP file, XML schema etc.) see [\[d2ref\]](#). For additional information see [\[d2supp\]](#).

(DATEX II) profile

simplified class model, still sufficient for specific needs. Profile is created from model by selecting only specific parts.

(DATEX II) extension

extended DATEX II model, adding data structures missing in standard model but needed for specific needs.

B level extension

Extension of DATEX II model, which is backward compatible with the original DATEX II model, even though it contains additional structures. [16157-1]

C level extension

Extension of DATEX II model, which is not backward compatible with the original DATEX II model.

publication

an element of DATEX II model, serving as an envelope for transmitted content of certain kind. Every sent DATEX II message contains exactly one element of publication type. There are various types of publications, for example *SituationPublication* [16157-3], *ElaboratedDataPublication* [16157-5], *PredefinedLocationsPublication* [16157-3] etc.

(W3C) XML schema

language, which allows definition of rules for a XML document structure.

XSD

XML document containing XML schema. Sometimes it is synonym for XML schema itself.

uri

uniform resource identifier, text, used as unique identifier. In this document, every format has it's own uri.

CONCEPTS

The described format allows the provision of floating car data (FCD) information. This includes travel times, traffic flow speeds, traffic levels and estimation of congestion lengths on a set of predefined locations.

2.1 Publications of type *ElaboratedDataPublication*

The format is based on the DATEX II publication type *ElaboratedDataPublication* [16157-5] and uses it to transfer information about **travel times**, **speeds** and **traffic levels** on a set of predefined locations. The information about **traffic levels**, **length of queues** with their **start and end** is added via the level B extension of the *TrafficStatus* class of the DATEX II model.

Information about **data quality** and **number of FCD vehicles** on the segment, including their type, is added via the level B extension of the *TravelTimeData* class of the DATEX II model.

```
1 <?xml version="1.0" encoding="utf-8"?>
2 <d2LogicalModel modelBaseVersion="2" xmlns="http://datex2.eu/schema/2/2_0" xmlns:xsi=
   ↳ "http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="http://datex2.eu/
   ↳ schema/2/2_0 ../schema/schema.xsd">
3   <exchange>
4     <supplierIdentification>
5       <country>cz</country>
6       <nationalIdentifier>ŘSD</nationalIdentifier>
7     </supplierIdentification>
8   </exchange>
9   <payloadPublication lang="cs" xsi:type="ElaboratedDataPublication">
10    <feedType>x-format:cz-ndic_d2-fcd-v1.0</feedType>
11    <publicationTime>2017-11-18T00:00:06+01:00</publicationTime>
12    <publicationCreator>
13      <country>cz</country>
14      <nationalIdentifier>ŘSD</nationalIdentifier>
15    </publicationCreator>
16    <periodDefault>300</periodDefault>
17    <timeDefault>2017-11-17T23:59:35+01:00</timeDefault>
18    <headerInformation>
19      <confidentiality>noRestriction</confidentiality>
20      <informationStatus>real</informationStatus>
21    </headerInformation>
```

FCD information for a particular route/segment is listed in 3 *elaboratedData* elements, since the *elaboratedData* element can only hold **one type of information at one time**:

- traffic density in the *TrafficStatus* class + traffic levels (by CZ) and congestions in its extension,
- traffic speed in class *TrafficSpeed* and
- travel time in class *TravelTimeData* + quality parameters in its extension.

The *elaboratedData* element does not have its own unique identifier and is uniquely determined by the publication time and the referenced location identifier.

The identification of the profile of the transport information provided is given in the *feedType* element referring to the specific documentation published in the NAP (national access point).

2.2 Calculated FCD information

The status information included in this profile is:

- traffic flow speed [km/h] per
 - actual conditions
 - in free flow conditions (freeFlow)
- travel times [s] per
 - actual conditions
 - in free flow conditions (freeFlow)
- actual delay [s] (as the difference between actual - free flow travel times)
- traffic density [-] (as determined by the *trafficStatusEnum* enumeration)

Other information included in this profile is:

- system response times [s] (as the difference between the input data availability time and the status information calculation time)

The status information included in this profile by extension of the *TrafficStatus* class is:

- traffic levels [-] (1-5 as used in the Czech Republic)
- queues on the segment, estimation of their length [m] and estimation of the start [%] and end [%] of the queue on the referenced segment

The common qualitative parameters included in this profile by extending the *TravelTimeData* class are:

- data confidence level [-]
- number of FCD vehicle probes on the segment [-]

Information included by the extension is shown in Figure 1.

The following articles are dedicated to the description of each status information.

2.2.1 Data Reliability Measures

The **Data Reliability Rate** characterizes the quality of the data sample and has values <0-1> where 1 means 100% reliable data. The quality of the data sample is evaluated by the age of the input positions used, their number, etc. The calculation formula is not part of this profile.

The data reliability rate is given for a segment in the *travelTimeDataExtension* in the *supplierCalculatedDataQuality* element and applies to all *elaboratedData* elements related to the segment.

```

<travelTimeDataExtension>
  <qualityParameters>
    <supplierCalculatedDataQuality>1</supplierCalculatedDataQuality>
  
```

The profile allows (but does not require) specifying the data reliability rate for each calculated value separately by the *supplierCalculatedDataQuality* attribute. If specified, this value takes precedence over the value at a more generic level.

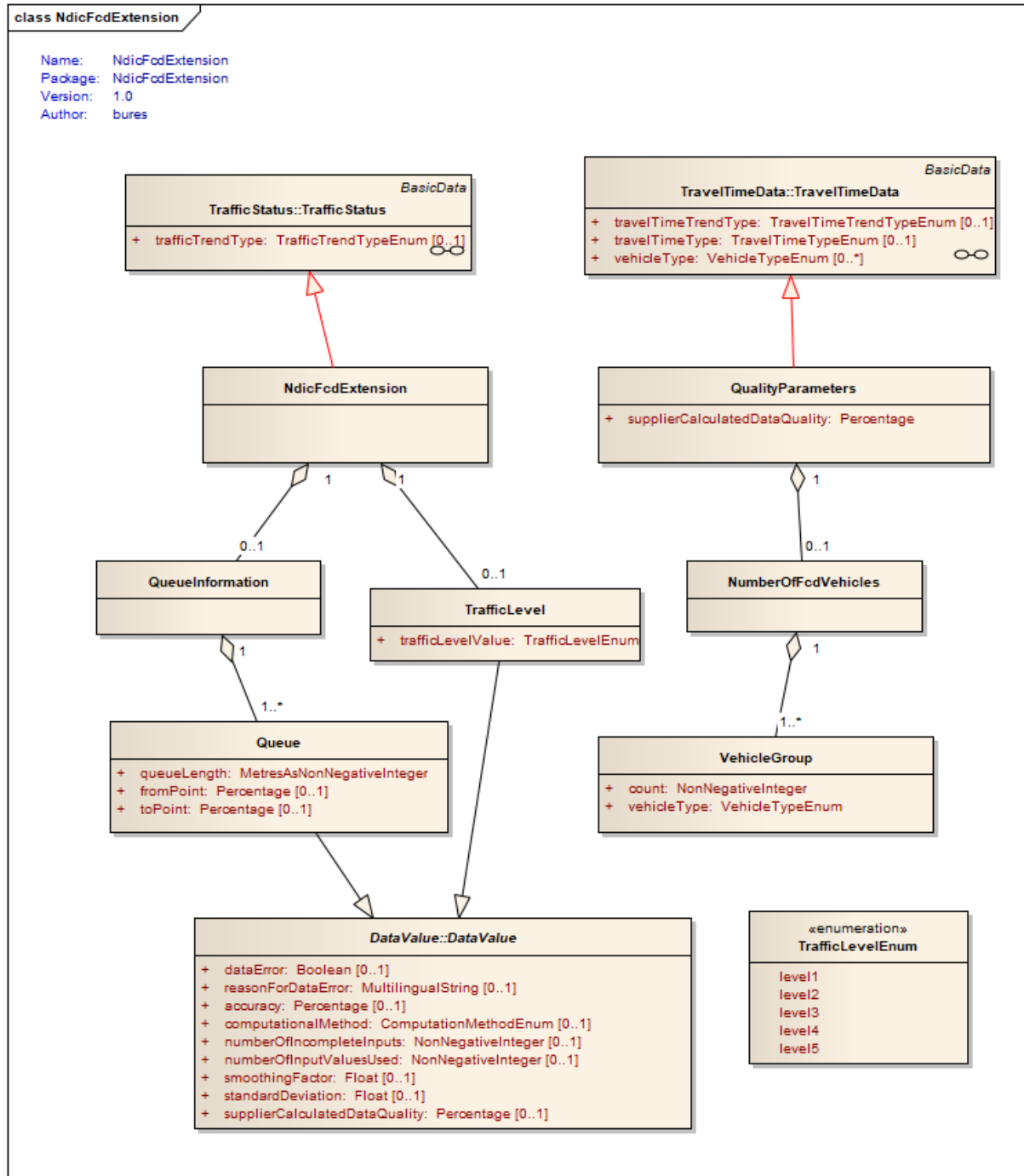


Fig. 1: TrafficStatus and TravelTimeData class extension.


```

92 <freeFlowSpeed supplierCalculatedDataQuality="0.8">
93   <speed>50</speed>
94 </freeFlowSpeed>

```

2.2.2 Actual number of floating vehicles

The actual **number of floating vehicles** from which the values were calculated is given in the *numberOfFcdVehicles* element of the *trafficStatusExtension* element and applies to all *elaboratedData* elements related to the segment.

The **number of floating vehicles** can be specified in more detail by vehicle category. The vehicle category is expressed by the value of the *vehicleType* element embedded in the *vehicleGroup* element. One *vehicleGroup* may be used for each value of *vehicleType*, with the value of *anyVehicle* being used to express the total number of vehicles of all categories.

```

99   <numberOfFcdVehicles>
100     <vehicleGroup>
101       <count>9</count>
102       <vehicleType>anyVehicle</vehicleType>
103     </vehicleGroup>
104   </numberOfFcdVehicles>

```

The profile allows (but does not require) specifying the total number of vehicle probes used for each calculated value separately by the *numberOfInputValuesUsed* attribute.

```

34 <trafficStatus numberOfInputValuesUsed="9" supplierCalculatedDataQuality="1">
35   <trafficStatusValue>congested</trafficStatusValue>
36 </trafficStatus>

```

Some statistics, such as *freeFlowSpeed* or *freeFlowTravelTime*, are independent of the number of floating vehicles. In this case, the value of the number of vehicles, if specified by the *numberOfInputValuesUsed* attribute, is set to 9999.

```

88 <freeFlowTravelTime numberOfInputValuesUsed="9999"
89   supplierCalculatedDataQuality="1">
90   <duration>259</duration>
91 </freeFlowTravelTime>

```

2.2.3 Traffic flow speed

The **traffic flow speed** [km/h] on the segment is given by two separate *elaboratedData* elements. The actual average speed *averageVehicleSpeed* in the measured time interval is given in the *TrafficSpeed* class of the *basicData* element.

```

67 <averageVehicleSpeed>
68   <speed>38</speed>
69 </averageVehicleSpeed>

```

The *freeFlowSpeed* [km/h] on the segment is specified in the *TravelTimeData* class in the *basicData* element.

```

92 <freeFlowSpeed supplierCalculatedDataQuality="0.8">
93   <speed>50</speed>
94 </freeFlowSpeed>

```

2.2.4 Travel time and actual delay

The **Travel Time** [s] is given in the *basicData* element with *xsi:type="TravelTimeData"*. There are 2 different types of travel times used in this profile:

- *travelTime*: actual average travel time.
- *freeFlowTravelTime*: free flow travel time

```

83     <vehicleType>anyVehicle</vehicleType>
84     <travelTime numberOfInputValuesUsed="9" supplierCalculatedDataQuality="1">
85       <duration>549</duration>
86     </travelTime>
87     <!-- historical data from fcd vehicles database -->
88     <freeFlowTravelTime numberOfInputValuesUsed="9999"
89     ↪supplierCalculatedDataQuality="1">
90       <duration>259</duration>
91     </freeFlowTravelTime>

```

The profile provides information about the travel time without distinguishing the vehicle type.

The **actual delay** is expressed implicitly as the difference between the values of the actual average travel time (*travelTime*) and the free flow travel time (*freeFlowTravelTime*).

2.2.5 Traffic levels

Traffic status is expressed by *TrafficStatusEnum* enumeration as defined in standard [16157-3] in table A.125.

```

34     <trafficStatus numberOfInputValuesUsed="9" supplierCalculatedDataQuality="1">
35       <trafficStatusValue>congested</trafficStatusValue>
36     </trafficStatus>

```

Table 1: Values of the “TrafficStatusEnum” enumeration.

value	Meaning	Definition
impossible	Traffic col-lapse	Vehicles on the roads are stationary or in convoys only moving very slowly, the average speed is very low.
congested	Traffic congestion	Vehicle queues are formed, traffic is not flowing smoothly, average speed is significantly reduced, intersection passage is impaired.
heavy	Heavy traffic	Streams of vehicles form, traffic is flowing, but speed is below the maximum speed limit.
freeFlow	Flowing traffic	Traffic is flowing in that direction. Traffic is moving at or above the speed limit and drivers are free to change lanes.
unknown	Unknown	Traffic conditions are unknown.

The *TrafficStatus* class is extended (level B extension) to explicitly specify the traffic levels as defined and used in the CR. The traffic levels (1-5) are specified in the *TrafficLevel* class using the phrases “level1” to “level5”, in the *trafficLevelValue* element.

```

40     <trafficLevel>
41       <trafficLevelValue>level4</trafficLevelValue>
42     </trafficLevel>

```

The *TrafficLevel* is calculated as the % difference between the current traffic flow speed and the traffic flow speed on the free segment.

Table 2: Calculation in the definition of traffic stages in the FCD publication

Degree	Description	% of difference
1	single vehicle operation only, driving is smooth	<85 ‰, 100 ‰)
2	small groups of vehicles, smooth running, passing intersections without problems	<75 ‰, 85 ‰)
3	vehicle streams are forming, traffic is smooth but speed is below the maximum speed limit	<50 ‰, 75 ‰)
4	vehicle queues are forming, traffic is not flowing, average speed is significantly reduced, passing the intersections is impaired	<20 ‰, 50 ‰)
5	traffic collapse - vehicles are stationary on the roads or moving very slowly in convoys, average speed is very low	<0 ‰, 20 ‰)

2.2.6 Congestions

The **congestions** on a segment are expressed implicitly by the traffic status “level4” and “level5” and by the presence of the *queueInformation* element, which was added to the *TrafficStatus* class via the DATEX II type B extension.

Segments are internally divided into smaller parts (sections of the Global Network, GN road network). The presence of congestion is evaluated with respect to GN sections as a decrease in the travelled speed of this section **below 30 ‰ of the free-flow speed**.

The *queueInformation* element contains aggregated values from individual congested sections of the segment. If none of the sections of the segment were congested, the *queueInformation* element is not present. The element is present only once per segment, the values of its attributes are adapted to this (for more, see the description of the attributes and the image below).

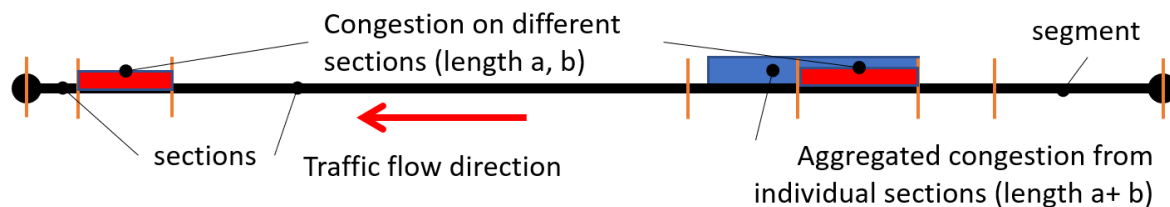


Fig. 2: Representation of the aggregation of individual congestions on a segment

```

43 <queueInformation>
44 <queue>
45 <queueLength>148</queueLength>
46 <!-- start of the queue (tail) at the 0.4 of the length of the segment -->
47 <fromPoint>0.4</fromPoint>
48 <!-- end of the queue (head) at the 0.9 of the length of the segment -->
49 <toPoint>0.9</toPoint>
50 </queue>
51 </queueInformation>

```

The length of a queue on a referenced segment is expressed in the *queueLength* element and is aggregated length of the traffic congestion for all sections of the segment [m]. i.e. the sum of the lengths of the **congested** Global Network sections (even non-contiguous) lying on the segment.

The start and end of the queue is given by the optional *fromPoint* and *toPoint* attributes, expressing the relative *coordinates* of the queue relative to the beginning segment in the direction of traffic. A queue, according to this

definition, starts (*fromPoint*) at the point where vehicles enter it and ends (*toPoint*) at the point where they leave it and continue on their route. See Figure 2.

In the case of multiple split congestions on a segment, for example, a congestion detected on GN sections 1 and 3, the *fromPoint* attribute only gives information about the 1st congestion and *toPoint* is calculated by using the *queueLength* value.

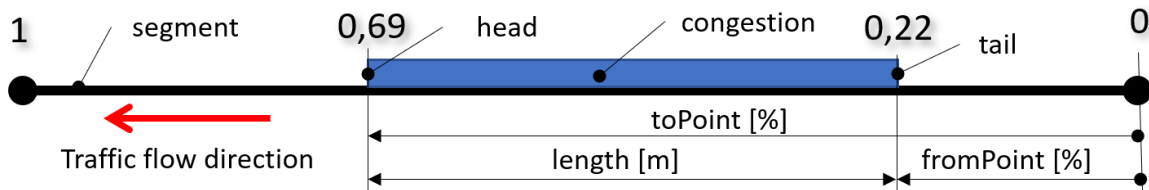


Fig. 3: Example of referencing the relative position of a queue on a segment.

2.3 Providing only currently available data

In case of unavailability of input data for the status information calculation on a given segment, only qualitative parameters are given for this segment and, and if available, also historical data in the *basicData* element of the *TravelTimeData* class.

Thus, a segment for which actual status data exists is covered by 3 *elaboratedData* elements, while a segment for which no actual data exists is covered only by one element.

TEMPORAL VALIDITY

The publication is periodically updated, the value in the *periodDefault* element specifies the normal length of the update interval in seconds (normally 5 minutes).

The value of *publicationTime* specifies when the publication was created, but **not the reference time of the calculation**.

The *timeDefault* value specifies the **common reference time of all record values** and is the end time of a particular measurement and calculation period.

No other determination of data validity is used.

```
11 <publicationTime>2017-11-18T00:00:06+01:00</publicationTime>
12 <publicationCreator>
13   <country>cz</country>
14   <nationalIdentifier>ŘSD</nationalIdentifier>
15 </publicationCreator>
16 <periodDefault>300</periodDefault>
17 <timeDefault>2017-11-17T23:59:35+01:00</timeDefault>
```

3.1 FCD input data age

Each individual record is described by a *basicData* element, and in its *measurementOrCalculationTime* element gives the time last input information used to calculate status values. The update takes place regularly at an interval of 1 minute. At that time, the routes of all vehicles with respect to travelled sections (parts of segments) of the monitored network are evaluated. The last update of the segment status occurs when the last vehicle leaves the segment.

The age of the input data used for each record can be calculated as the difference between the *timeDefault* of the entire publication and the *measurementOrCalculationTime* value of that record. The age of the input data can be used to determine **system response time**.

The status values of individual records are calculated using both actual input data and historical data. The status values of a record without new input data from the FCD vehicle probes may still change in subsequent publications as historical data is gradually taken into account. The calculation algorithm used is beyond the scope of this documentation.

In case of insufficient underlying data for the calculation of the statuses (according to the calculation algorithm), elements containing information on **actual** speed, traffic stages and travel time are omitted.

```
16 <periodDefault>300</periodDefault>
17 <timeDefault>2017-11-17T23:59:35+01:00</timeDefault>
18 <headerInformation>
19   <confidentiality>noRestriction</confidentiality>
20   <informationStatus>real</informationStatus>
21 </headerInformation>
22 <!-- information from one segment -->
```

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```
23 <!-- traffic status element -->
24 <elaboratedData>
25   <source>
26     <sourceIdentification>ndic-secid-1-01</sourceIdentification>
27   </source>
28   <basicData xsi:type="TrafficStatus">
29     <measurementOrCalculationTime>2017-11-17T23:57:35+01:00</
    measurementOrCalculationTime>
```

The degree to which historical data is used in the calculation of the state value is reflected in the value of *supplier-CalculatedDataQuality*, among other things.

LOCATION DESCRIPTION

FCD data are measured at predefined road segments. The segment length can be given by ALERT-C location points distance, sections of the road network used (GlobalNetwork), or requirements based on the physical layout of the road.

The location of FCD segment is expressed by a reference in a *pertinentLocation* element to a set of predefined locations in a *PredefinedLocationsPublication* [16157-3] associated with this information source.

```
30      <pertinentLocation xsi:type="LocationByReference">
31          <predefinedLocationReference id="6D476A44-AA44-4947-AA4B-7C4DD79BB67E"
32      ↪targetClass="PredefinedLocation" version="0"/>
      </pertinentLocation>
```

In addition to describing the location with the *PredefinedLocation* element, the format optionally allows the location to be described with an additional reference, the provider's internal identifier, in the *source element*.

```
25      <source>
26          <sourceIdentification>ndic-secid-1-01</sourceIdentification>
27      </source>
```

Location description methods are beyond the scope of this document and are described elsewhere.

ANNEX A: EXAMPLES OF MESSAGES

5.1 samples/fcd-compatibility-mode.xml - FCD data on 1 segment

The sample contains FCD information for one physical location (route with identifier *6D476A44-AA44-4947-AA4B-7C4DD79BB67E*).

The routes are defined in another publication of type *PredefinedLocationsPublication*, which is not described in detail here.

```
<?xml version="1.0" encoding="utf-8"?>
<d2LogicalModel modelBaseVersion="2" xmlns="http://datex2.eu/schema/2/2_0" xmlns:xsi=
  ↪ "http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="http://datex2.eu/
  ↪ schema/2/2_0 ../schema/schema.xsd">
  <exchange>
    <supplierIdentification>
      <country>cz</country>
      <nationalIdentifier>ŘSD</nationalIdentifier>
    </supplierIdentification>
  </exchange>
  <payloadPublication lang="cs" xsi:type="ElaboratedDataPublication">
    <feedType>x-format:cz-ndic_d2-fcd-v1.0</feedType>
    <publicationTime>2017-11-18T00:00:06+01:00</publicationTime>
    <publicationCreator>
      <country>cz</country>
      <nationalIdentifier>ŘSD</nationalIdentifier>
    </publicationCreator>
    <periodDefault>300</periodDefault>
    <timeDefault>2017-11-17T23:59:35+01:00</timeDefault>
    <headerInformation>
      <confidentiality>noRestriction</confidentiality>
      <informationStatus>real</informationStatus>
    </headerInformation>
    <!-- information from one segment -->
    <!-- traffic status element -->
    <elaboratedData>
      <source>
        <sourceIdentification>ndic-secid-1-01</sourceIdentification>
      </source>
      <basicData xsi:type="TrafficStatus">
        <measurementOrCalculationTime>2017-11-17T23:57:35+01:00</
  ↪ measurementOrCalculationTime>
        <pertinentLocation xsi:type="LocationByReference">
          <predefinedLocationReference id="6D476A44-AA44-4947-AA4B-7C4DD79BB67E"
  ↪ ↪ targetClass="PredefinedLocation" version="0"/>
        </pertinentLocation>
```

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```

<!-- traffic status information -->
<trafficStatus numberOfInputValuesUsed="9" supplierCalculatedDataQuality="1">
  <trafficStatusValue>congested</trafficStatusValue>
</trafficStatus>
<!-- extension: traffic level and queue information -->
<trafficStatusExtension>
  <ndicFcdExtension>
    <trafficLevel>
      <trafficLevelValue>level4</trafficLevelValue>
    </trafficLevel>
    <queueInformation>
      <queue>
        <queueLength>148</queueLength>
        <!-- start of the queue (tail) at the 0.4 of the length of the
↪ segment -->
        <fromPoint>0.4</fromPoint>
        <!-- end of the queue (head) at the 0.9 of the length of the segment -
↪ ->
        <toPoint>0.9</toPoint>
      </queue>
    </queueInformation>
  </ndicFcdExtension>
</trafficStatusExtension>
</basicData>
</elaboratedData>
<!-- speed information element -->
<elaboratedData>
  <source>
    <sourceIdentification>ndic-secid-1-01</sourceIdentification>
  </source>
  <basicData xsi:type="TrafficSpeed">
    <measurementOrCalculationTime>2017-11-17T23:57:35+01:00</
↪ measurementOrCalculationTime>
    <pertinentLocation xsi:type="LocationByReference">
      <predefinedLocationReference id="6D476A44-AA44-4947-AA4B-7C4DD79BB67E"
↪
↪ targetClass="PredefinedLocation" version="0"/>
    </pertinentLocation>
    <!-- average vehicle speed information -->
    <averageVehicleSpeed>
      <speed>38</speed>
    </averageVehicleSpeed>
  </basicData>
</elaboratedData>
<!-- travel time element -->
<elaboratedData>
  <source>
    <sourceIdentification>ndic-secid-1-01</sourceIdentification>
  </source>
  <basicData xsi:type="TravelTimeData">
    <measurementOrCalculationTime>2017-11-17T23:57:35+01:00</
↪ measurementOrCalculationTime>
    <pertinentLocation xsi:type="LocationByReference">
      <predefinedLocationReference id="6D476A44-AA44-4947-AA4B-7C4DD79BB67E"
↪
↪ targetClass="PredefinedLocation" version="0"/>
    </pertinentLocation>
    <!-- travel time information -->

```

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```

<vehicleType>anyVehicle</vehicleType>
<travelTime numberOfInputValuesUsed="9" supplierCalculatedDataQuality="1">
  <duration>549</duration>
</travelTime>
<!-- historical data from fcd vehicles database -->
<freeFlowTravelTime numberOfInputValuesUsed="9999"
↪supplierCalculatedDataQuality="1">
  <duration>259</duration>
</freeFlowTravelTime>
<!-- free flow vehicle speed information -->
<freeFlowSpeed supplierCalculatedDataQuality="0.8">
  <speed>50</speed>
</freeFlowSpeed>
<!-- extension: quality parameters and number of vehicles on the segment -->
<travelTimeDataExtension>
  <qualityParameters>
    <supplierCalculatedDataQuality>1</supplierCalculatedDataQuality>
    <numberOfFcdVehicles>
      <vehicleGroup>
        <count>9</count>
        <vehicleType>anyVehicle</vehicleType>
      </vehicleGroup>
    </numberOfFcdVehicles>
  </qualityParameters>
</travelTimeDataExtension>
</basicData>
</elaboratedData>
<!-- information from another segment follows -->
</payloadPublication>
</d2LogicalModel>

```

5.1.1 Root element and the publication envelope

The root element *d2LogicalModel* declares that this is a message using DATEX II.

The *exchange* element contains the provider identifier.

The *payloadPublication* element is an envelope for the information itself (payload), containing primarily *elaboratedData* elements, of which there may be many (seven in this example).

The *xsi:type="ElaboratedDataPublication"* attribute in the *payloadPublication* element specifies that this envelope is of type *ElaboratedData*.

The *feedType* element specifies the type of publication, by the uri of the schema.

The *publicationTime* element specifies the time of the publication.

The *publicationCreator* element identifies the creator of the publication.

The *timeDefault* element indicates the time of validity of the calculated traffic statuses, this is the time of calculation of all FCD values given in the report.

5.1.2 General characteristics of the *elaboratedData* elements

The following three *elaboratedData* elements list the individual characteristics of the FCD measurements. However, a few characteristics are similar/same for all *elaboratedData* elements, they are listed below.

Location reference

The *id*="6D476A44-AA44-4947-AA4B-7C4DD79BB67E" attribute in the *predefinedLocationReference* elements of each *elaboratedData* record refers to a predefined segment (location) with a given identifier, and the *version*="0" attribute specifies which version of the location reference it is.

Temporal validity

The start of validity of the given status values is specified together for all records in the *timeDefault* element (2017-11-17T23:59:35+01:00).

The last time of occurrence of vehicles, data sources, (2017-11-17T23:57:35+01:00) on the measured segment is given in the *measurementOrCalculationTime* elements. This value may be different for each segment.

Measurement quality

The measurement given in the *qualityParameterssupplierCalculatedDataQuality* elements and the optional *supplierCalculatedDataQuality* attributes of the calculated values are set to 1. It means "reliable". This value should be the same for all actual measurement values on the segment.

Number of vehicles on the segment

The element *numberOfFcdVehicles* and optional attributes *numberOfInputValuesUsed* are set to 9. It means there were 9 FCD vehicle probes on the given segment (route) at the time of measurement (300 s). This value should be the same for all the actual measurement values on a given segment.

Source identification

The optional *sourceIdentification* element in *elaboratedData* elements may provide an alternative route identification. The value is used to facilitate identification in addition to the location identifier.

5.1.3 Description of traffic levels and queues

The first element *elaboratedData* describes the calculated traffic status and its statistical data.

The nested element *basicData* indicates by its attribute *xsi:type*="TrafficStatus" that the values shall be interpreted as "traffic density".

The *trafficStatus* element indicates the level of traffic on a four-level scale, as "congested".

The following portions of the *elaboratedData* element are Level B extensions of the DATEX II model.

Traffic levels

The *trafficLevel* element lists the level of traffic in the *level* sub-element by enumeration type, in this case with a value of *level4*, i.e. level 4 (equivalent of congested).

This value also indicates the presence of a traffic jam.

Queue length

The queue information is encapsulated in the *queueInformation* element. Individual queues are listed in the *queue* element. In the *queueLength* sub-element the actual value of the length of one queue on a given route (place) at a given time = 148 m.

The sub-elements *fromPoint* and *toPoint* give the start (0.4 of the segment length) and end (0.9) of the segment length) of the queue in relative coordinates.

5.1.4 Description of the measured speed

The second element *elaboratedData* describes the computed traffic flow speed with its statistical data.

The nested element *basicData* indicates by its attribute *xsi:type="TrafficSpeed"* that the values shall be interpreted as "traffic speed".

Traffic speed

The *averageVehicleSpeed* element gives the actual average traffic speed of 38 km/h.

5.1.5 Description of travel time and speed

The third element *elaboratedData* describes the calculated actual and historical travel times and speed of the continuous traffic flow with their statistical data.

The nested element *basicData* indicates by its attribute *xsi:type="TravelTimeData"* that the values shall be interpreted as "travel times".

The *vehicleType* element indicates that the following values apply to all vehicles *anyVehicle*.

Travel time

The *travelTime* element gives the average actual travel time of 549 s.

The *freeFlowTravelTime* element gives the average travel time in continuous operation as 259 s.

The *normallyExpectedTravelTime* element gives the average travel time at a given location at a given time as 348 s, so this is an abnormal condition.

Speed

The *freeFlowSpeed* element gives the average speed in in freely flowing traffic at 50 km/h.

5.2 samples/fcd-compatibility-mode-no-data.xml - no FCD data on 1 segment

The sample contains FCD information for the same location as the previous sample. The difference between the samples is the **unavailability of data to calculate status information**.

```
<?xml version="1.0" encoding="utf-8"?>
<d2LogicalModel modelBaseVersion="2" xmlns="http://datex2.eu/schema/2/2_0" xmlns:xsi=
  ↪ "http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="http://datex2.eu/
  ↪ schema/2/2_0 ../schema/schema.xsd">
  <exchange>
    <supplierIdentification>
      <country>cz</country>
      <nationalIdentifier>ŘSD</nationalIdentifier>
    </supplierIdentification>
  </exchange>
  <payloadPublication lang="cs" xsi:type="ElaboratedDataPublication">
    <feedType>x-format:cz-ndic_d2-fcd-v1.0</feedType>
    <publicationTime>2017-11-18T00:00:06+01:00</publicationTime>
    <publicationCreator>
      <country>cz</country>
      <nationalIdentifier>ŘSD</nationalIdentifier>
    </publicationCreator>
    <periodDefault>300</periodDefault>
    <timeDefault>2017-11-17T23:59:35+01:00</timeDefault>
    <headerInformation>
      <confidentiality>noRestriction</confidentiality>
      <informationStatus>real</informationStatus>
    </headerInformation>
    <!-- start of information from one segment -->
    <!-- NO status information element HERE due to 0 fcd vehicles, element_
  ↪ elaboratedData is missing -->
    <!-- NO speed information element HERE due to 0 fcd vehicles, element_
  ↪ elaboratedData is missing -->
    <elaboratedData>
      <source>
        <sourceIdentification>ndic-secid-1-01</sourceIdentification>
      </source>
      <basicData xsi:type="TravelTimeData">
        <measurementOrCalculationTime>2017-11-17T22:57:35+01:00</
  ↪ measurementOrCalculationTime>
        <pertinentLocation xsi:type="LocationByReference">
          <predefinedLocationReference id="6D476A44-AA44-4947-AA4B-7C4DD79BB67E"
  ↪ targetClass="PredefinedLocation" version="0"/>
        </pertinentLocation>
        <!-- NO travel time information element HERE due to 0 fcd vehicles, element_
  ↪ travelTime is missing -->
        <!-- historical data from fcd vehicles database -->
        <freeFlowTravelTime>
          <duration>259</duration>
        </freeFlowTravelTime>
        <!-- free flow vehicle speed information-->
        <freeFlowSpeed>
          <speed>50</speed>
        </freeFlowSpeed>
        <!-- extension: quality parameters and number of vehicles on the segment -->
```

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```

    <travelTimeDataExtension>
      <qualityParameters>
        <supplierCalculatedDataQuality>0</supplierCalculatedDataQuality>
        <numberOfFcdVehicles>
          <vehicleGroup>
            <count>0</count>
            <vehicleType>anyVehicle</vehicleType>
          </vehicleGroup>
        </numberOfFcdVehicles>
      </qualityParameters>
    </travelTimeDataExtension>
  </basicData>
</elaboratedData>
  <!-- information from another segment follows -->
</payloadPublication>
</d2LogicalModel>

```

In the sections below, we only list the differences from the previous sample.

5.2.1 General characteristics of *elaboratedData* elements

The following three *elaboratedData* elements list the individual characteristics of the FCD measurements. However, a few characteristics are similar/same for all *elaboratedData* elements, they are listed below.

Temporal validity

The start of validity of the given status values is specified together for all records in the *timeDefault* element is 2017-11-17T23:59:35+01:00.

The last time of occurrence of vehicles, data sources, on the measured segment listed in the *measurementOrCalculationTime* element is 2017-11-17T22:57:35+01:00, about an hour ago.

Measurement quality

Value in *qualityParameterssupplierCalculatedDataQuality* and optional *supplierCalculatedDataQuality* is 0. It means “not-reliable”.

Number of vehicles on the segment

Value in *numberOfFcdVehicles* and optional attribute *numberOfInputValuesUsed* is 0. It means there were NO FCD vehicle probes on the given segment at the time of measurement (300 s).

5.2.2 Description of traffic levels and queues

The first element *elaboratedData* describes the calculated traffic status, traffic level and queues.

The related element *elaboratedData* is not provided.

5.2.3 Description of the measured speed

The second element *elaboratedData* describes the calculated traffic flow speed.

The related element *elaboratedData* is not provided.

5.2.4 Description of travel time and speed

The third element *elaboratedData* describes the calculated actual and historical travel times and speed of the continuous traffic flow with their statistical data.

Travel time

Only the *freeFlowTravelTime* element is given, indicating an average travel time in a free flow traffic conditions of 259 s.

Speed

Only the *freeFlowSpeed* element is given, giving an average speed in a free flow traffic conditions at 50 km/h.

APPENDIX B: W3C XML SCHEMA

XML structure of a message is defined by following W3C XML schema:

6.1 *schema.xsd*

```

1 <?xml version="1.0" encoding="utf-8" standalone="no"?>
2 <xs:schema elementFormDefault="qualified" attributeFormDefault="unqualified"
  ↳ xmlns:D2LogicalModel="http://datex2.eu/schema/2/2_0" version="2.3" targetNamespace=
  ↳ "http://datex2.eu/schema/2/2_0" xmlns:xs="http://www.w3.org/2001/XMLSchema">
3   <xs:complexType name="_ExtensionType">
4     <xs:sequence>
5       <xs:any namespace="##any" processContents="lax" minOccurs="0" maxOccurs=
  ↳ "unbounded" />
6     </xs:sequence>
7   </xs:complexType>
8   <xs:complexType name="_PredefinedLocationVersionedReference">
9     <xs:complexContent>
10      <xs:extension base="D2LogicalModel:VersionedReference">
11        <xs:attribute name="targetClass" use="required" fixed="PredefinedLocation" />
12      </xs:extension>
13    </xs:complexContent>
14  </xs:complexType>
15  <xs:complexType name="_TrafficStatusExtensionType">
16    <xs:sequence>
17      <xs:element name="ndicFcdExtension" type="D2LogicalModel:NdicFcdExtension"
  ↳ minOccurs="0" />
18      <xs:any namespace="##other" processContents="lax" minOccurs="0" maxOccurs=
  ↳ "unbounded" />
19    </xs:sequence>
20  </xs:complexType>
21  <xs:complexType name="_TravelTimeDataExtensionType">
22    <xs:sequence>
23      <xs:element name="qualityParameters" type="D2LogicalModel:QualityParameters"
  ↳ minOccurs="0" />
24      <xs:any namespace="##other" processContents="lax" minOccurs="0" maxOccurs=
  ↳ "unbounded" />
25    </xs:sequence>
26  </xs:complexType>
27  <xs:simpleType name="AreaOfInterestEnum">
28    <xs:restriction base="xs:string">
29      <xs:enumeration value="continentWide" />
30      <xs:enumeration value="national" />
31      <xs:enumeration value="neighbouringCountries" />
32      <xs:enumeration value="notSpecified" />

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```

33     <xs:enumeration value="regional" />
34   </xs:restriction>
35 </xs:simpleType>
36 <xs:complexType name="BasicData" abstract="true">
37   <xs:sequence>
38     <xs:element name="measurementOrCalculationTime" type="D2LogicalModel:DateTime"
↪ minOccurs="0" maxOccurs="1" />
39     <xs:element name="pertinentLocation" type="D2LogicalModel:GroupOfLocations"
↪ minOccurs="0" />
40     <xs:element name="basicDataExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
41   </xs:sequence>
42 </xs:complexType>
43 <xs:simpleType name="Boolean">
44   <xs:restriction base="xs:boolean" />
45 </xs:simpleType>
46 <xs:simpleType name="ComputationMethodEnum">
47   <xs:restriction base="xs:string">
48     <xs:enumeration value="arithmeticAverageOfSamplesBasedOnAFixedNumberOfSamples" /
↪ >
49     <xs:enumeration value="arithmeticAverageOfSamplesInATimePeriod" />
50     <xs:enumeration value="harmonicAverageOfSamplesInATimePeriod" />
51     <xs:enumeration value="medianOfSamplesInATimePeriod" />
52     <xs:enumeration value="movingAverageOfSamples" />
53   </xs:restriction>
54 </xs:simpleType>
55 <xs:simpleType name="ConfidentialityValueEnum">
56   <xs:restriction base="xs:string">
57     <xs:enumeration value="internalUse" />
58     <xs:enumeration value="noRestriction" />
59     <xs:enumeration value="restrictedToAuthorities" />
60     <xs:enumeration value="restrictedToAuthoritiesAndTrafficOperators" />
61     <xs:enumeration value="restrictedToAuthoritiesTrafficOperatorsAndPublishers" />
62     <xs:enumeration value="restrictedToAuthoritiesTrafficOperatorsAndVms" />
63   </xs:restriction>
64 </xs:simpleType>
65 <xs:simpleType name="CountryEnum">
66   <xs:restriction base="xs:string">
67     <xs:enumeration value="at" />
68     <xs:enumeration value="be" />
69     <xs:enumeration value="bg" />
70     <xs:enumeration value="ch" />
71     <xs:enumeration value="cs" />
72     <xs:enumeration value="cy" />
73     <xs:enumeration value="cz" />
74     <xs:enumeration value="de" />
75     <xs:enumeration value="dk" />
76     <xs:enumeration value="ee" />
77     <xs:enumeration value="es" />
78     <xs:enumeration value="fi" />
79     <xs:enumeration value="fo" />
80     <xs:enumeration value="fr" />
81     <xs:enumeration value="gb" />
82     <xs:enumeration value="gg" />
83     <xs:enumeration value="gi" />
84     <xs:enumeration value="gr" />

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```

85     <xs:enumeration value="hr" />
86     <xs:enumeration value="hu" />
87     <xs:enumeration value="ie" />
88     <xs:enumeration value="im" />
89     <xs:enumeration value="is" />
90     <xs:enumeration value="it" />
91     <xs:enumeration value="je" />
92     <xs:enumeration value="li" />
93     <xs:enumeration value="lt" />
94     <xs:enumeration value="lu" />
95     <xs:enumeration value="lv" />
96     <xs:enumeration value="ma" />
97     <xs:enumeration value="mc" />
98     <xs:enumeration value="mk" />
99     <xs:enumeration value="mt" />
100    <xs:enumeration value="nl" />
101    <xs:enumeration value="no" />
102    <xs:enumeration value="pl" />
103    <xs:enumeration value="pt" />
104    <xs:enumeration value="ro" />
105    <xs:enumeration value="se" />
106    <xs:enumeration value="si" />
107    <xs:enumeration value="sk" />
108    <xs:enumeration value="sm" />
109    <xs:enumeration value="tr" />
110    <xs:enumeration value="va" />
111    <xs:enumeration value="other" />
112  </xs:restriction>
113 </xs:simpleType>
114 <xs:element name="d2LogicalModel" type="D2LogicalModel:D2LogicalModel" />
115 <xs:complexType name="D2LogicalModel">
116   <xs:sequence>
117     <xs:element name="exchange" type="D2LogicalModel:Exchange" />
118     <xs:element name="payloadPublication" type="D2LogicalModel:PayloadPublication"
119     ↪ minOccurs="0" />
120     <xs:element name="d2LogicalModelExtension" type="D2LogicalModel:_ExtensionType"
121     ↪ minOccurs="0" />
122   </xs:sequence>
123   <xs:attribute name="modelBaseVersion" use="required" fixed="2" />
124 </xs:complexType>
125 <xs:complexType name="DataValue" abstract="true">
126   <xs:sequence>
127     <xs:element name="dataError" type="D2LogicalModel:Boolean" minOccurs="0"
128     ↪ maxOccurs="1" />
129     <xs:element name="reasonForDataError" type="D2LogicalModel:MultilingualString"
130     ↪ minOccurs="0" maxOccurs="1" />
131     <xs:element name="dataValueExtension" type="D2LogicalModel:_ExtensionType"
132     ↪ minOccurs="0" />
133   </xs:sequence>
134   <xs:attribute name="accuracy" type="D2LogicalModel:Percentage" use="optional" />
135   <xs:attribute name="computationalMethod" type=
136   ↪ "D2LogicalModel:ComputationMethodEnum" use="optional" />
137   <xs:attribute name="numberOfIncompleteInputs" type=
138   ↪ "D2LogicalModel:NonNegativeInteger" use="optional" />
139   <xs:attribute name="numberOfInputValuesUsed" type=
140   ↪ "D2LogicalModel:NonNegativeInteger" use="optional" />

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```

133     <xs:attribute name="smoothingFactor" type="D2LogicalModel:Float" use="optional" />
134     <xs:attribute name="standardDeviation" type="D2LogicalModel:Float" use="optional" />
135     </xs:complexType>
136     <xs:simpleType name="DateTime">
137       <xs:restriction base="xs:dateTime" />
138     </xs:simpleType>
139     <xs:complexType name="DurationValue">
140       <xs:complexContent>
141         <xs:extension base="D2LogicalModel:DataValue">
142           <xs:sequence>
143             <xs:element name="duration" type="D2LogicalModel:Seconds" minOccurs="1"
144             ↪ maxOccurs="1" />
145             <xs:element name="durationValueExtension" type="D2LogicalModel:_
146             ↪ ExtensionType" minOccurs="0" />
147           </xs:sequence>
148         </xs:extension>
149       </xs:complexContent>
150     </xs:complexType>
151     <xs:complexType name="ElaboratedData">
152       <xs:sequence>
153         <xs:element name="source" type="D2LogicalModel:Source" minOccurs="0" />
154         <xs:element name="elaboratedDataFault" type="D2LogicalModel:ElaboratedDataFault
155         ↪ " minOccurs="0" maxOccurs="unbounded" />
156         <xs:element name="basicData" type="D2LogicalModel:BasicData" minOccurs="0" />
157         <xs:element name="elaboratedDataExtension" type="D2LogicalModel:_ExtensionType"
158         ↪ minOccurs="0" />
159       </xs:sequence>
160     </xs:complexType>
161     <xs:complexType name="ElaboratedDataFault">
162       <xs:complexContent>
163         <xs:extension base="D2LogicalModel:Fault">
164           <xs:sequence>
165             <xs:element name="elaboratedDataFault" type=
166             ↪ "D2LogicalModel:ElaboratedDataFaultEnum" minOccurs="1" maxOccurs="1" />
167             <xs:element name="elaboratedDataFaultExtension" type="D2LogicalModel:_
168             ↪ ExtensionType" minOccurs="0" />
169           </xs:sequence>
170         </xs:extension>
171       </xs:complexContent>
172     </xs:complexType>
173     <xs:simpleType name="ElaboratedDataFaultEnum">
174       <xs:restriction base="xs:string">
175         <xs:enumeration value="intermittentDataValues" />
176         <xs:enumeration value="noDataValuesAvailable" />
177         <xs:enumeration value="spuriousUnreliableDataValues" />
178         <xs:enumeration value="unspecifiedOrUnknownFault" />
179         <xs:enumeration value="other" />
180       </xs:restriction>
181     </xs:simpleType>
182     <xs:complexType name="ElaboratedDataPublication">
183       <xs:complexContent>
184         <xs:extension base="D2LogicalModel:PayloadPublication">
185           <xs:sequence>

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```

181     <xs:element name="periodDefault" type="D2LogicalModel:Seconds" minOccurs="0"
↪ " maxOccurs="1" />
182     <xs:element name="timeDefault" type="D2LogicalModel:DateTime" minOccurs="0"
↪ maxOccurs="1" />
183     <xs:element name="headerInformation" type="D2LogicalModel:HeaderInformation"
↪ " />
184     <xs:element name="elaboratedData" type="D2LogicalModel:ElaboratedData"
↪ maxOccurs="unbounded" />
185     <xs:element name="elaboratedDataPublicationExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
186     </xs:sequence>
187     </xs:extension>
188     </xs:complexContent>
189   </xs:complexType>
190   <xs:complexType name="Exchange">
191     <xs:sequence>
192       <xs:element name="supplierIdentification" type=
↪ "D2LogicalModel:InternationalIdentifier" />
193       <xs:element name="exchangeExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
194     </xs:sequence>
195   </xs:complexType>
196   <xs:complexType name="Fault">
197     <xs:sequence>
198       <xs:element name="faultIdentifier" type="D2LogicalModel:String" minOccurs="0"
↪ maxOccurs="1" />
199       <xs:element name="faultDescription" type="D2LogicalModel:String" minOccurs="0"
↪ maxOccurs="1" />
200       <xs:element name="faultCreationTime" type="D2LogicalModel:DateTime" minOccurs="0"
↪ " maxOccurs="1" />
201       <xs:element name="faultLastUpdateTime" type="D2LogicalModel:DateTime" minOccurs=
↪ "1" maxOccurs="1" />
202       <xs:element name="faultSeverity" type="D2LogicalModel:FaultSeverityEnum"
↪ minOccurs="0" maxOccurs="1" />
203       <xs:element name="faultExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
204     </xs:sequence>
205   </xs:complexType>
206   <xs:simpleType name="FaultSeverityEnum">
207     <xs:restriction base="xs:string">
208       <xs:enumeration value="low" />
209       <xs:enumeration value="medium" />
210       <xs:enumeration value="high" />
211       <xs:enumeration value="unknown" />
212     </xs:restriction>
213   </xs:simpleType>
214   <xs:simpleType name="Float">
215     <xs:restriction base="xs:float" />
216   </xs:simpleType>
217   <xs:complexType name="GroupOfLocations" abstract="true">
218     <xs:sequence>
219       <xs:element name="groupOfLocationsExtension" type="D2LogicalModel:_ExtensionType"
↪ " minOccurs="0" />
220     </xs:sequence>
221   </xs:complexType>
222   <xs:complexType name="HeaderInformation">

```

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```

223 <xs:sequence>
224   <xs:element name="areaOfInterest" type="D2LogicalModel:AreaOfInterestEnum"
↪ minOccurs="0" maxOccurs="1" />
225   <xs:element name="confidentiality" type="D2LogicalModel:ConfidentialityValueEnum
↪ " minOccurs="1" maxOccurs="1" />
226   <xs:element name="informationStatus" type="D2LogicalModel:InformationStatusEnum
↪ " minOccurs="1" maxOccurs="1" />
227   <xs:element name="headerInformationExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
228 </xs:sequence>
229 </xs:complexType>
230 <xs:simpleType name="InformationStatusEnum">
231   <xs:restriction base="xs:string">
232     <xs:enumeration value="real" />
233     <xs:enumeration value="securityExercise" />
234     <xs:enumeration value="technicalExercise" />
235     <xs:enumeration value="test" />
236   </xs:restriction>
237 </xs:simpleType>
238 <xs:complexType name="InternationalIdentifier">
239   <xs:sequence>
240     <xs:element name="country" type="D2LogicalModel:CountryEnum" minOccurs="1"
↪ maxOccurs="1" />
241     <xs:element name="nationalIdentifier" type="D2LogicalModel:String" minOccurs="1
↪ " maxOccurs="1" />
242     <xs:element name="internationalIdentifierExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
243   </xs:sequence>
244 </xs:complexType>
245 <xs:simpleType name="KilometresPerHour">
246   <xs:restriction base="D2LogicalModel:Float" />
247 </xs:simpleType>
248 <xs:simpleType name="Language">
249   <xs:restriction base="xs:language" />
250 </xs:simpleType>
251 <xs:complexType name="Location" abstract="true">
252   <xs:complexContent>
253     <xs:extension base="D2LogicalModel:GroupOfLocations">
254       <xs:sequence>
255         <xs:element name="locationExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
256       </xs:sequence>
257     </xs:extension>
258   </xs:complexContent>
259 </xs:complexType>
260 <xs:complexType name="LocationByReference">
261   <xs:complexContent>
262     <xs:extension base="D2LogicalModel:Location">
263       <xs:sequence>
264         <xs:element name="predefinedLocationReference" type="D2LogicalModel:_
↪ PredefinedLocationVersionedReference" minOccurs="1" maxOccurs="1" />
265         <xs:element name="locationByReferenceExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
266       </xs:sequence>
267     </xs:extension>
268   </xs:complexContent>

```

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```

269 </xs:complexType>
270 <xs:complexType name="MeasurementEquipmentFault">
271   <xs:complexContent>
272     <xs:extension base="D2LogicalModel:Fault">
273       <xs:sequence>
274         <xs:element name="measurementEquipmentFault" type=
275 ↪ "D2LogicalModel:MeasurementEquipmentFaultEnum" minOccurs="1" maxOccurs="1" />
276         <xs:element name="measurementEquipmentFaultExtension" type="D2LogicalModel:_
277 ↪ ExtensionType" minOccurs="0" />
278       </xs:sequence>
279     </xs:extension>
280   </xs:complexContent>
281 </xs:complexType>
282 <xs:simpleType name="MeasurementEquipmentFaultEnum">
283   <xs:restriction base="xs:string">
284     <xs:enumeration value="intermittentDataValues" />
285     <xs:enumeration value="noDataValuesAvailable" />
286     <xs:enumeration value="spuriousUnreliableDataValues" />
287     <xs:enumeration value="unspecifiedOrUnknownFault" />
288     <xs:enumeration value="other" />
289   </xs:restriction>
290 </xs:simpleType>
291 <xs:simpleType name="MetresAsNonNegativeInteger">
292   <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
293 </xs:simpleType>
294 <xs:complexType name="MultilingualString">
295   <xs:sequence>
296     <xs:element name="values">
297       <xs:complexType>
298         <xs:sequence>
299           <xs:element name="value" type="D2LogicalModel:MultilingualStringValue"
300 ↪ maxOccurs="unbounded" />
301         </xs:sequence>
302       </xs:complexType>
303     </xs:element>
304   </xs:sequence>
305 </xs:complexType>
306 <xs:complexType name="MultilingualStringValue">
307   <xs:simpleContent>
308     <xs:extension base="D2LogicalModel:MultilingualStringValueType">
309       <xs:attribute name="lang" type="xs:language" />
310     </xs:extension>
311   </xs:simpleContent>
312 </xs:complexType>
313 <xs:simpleType name="MultilingualStringValueType">
314   <xs:restriction base="xs:string">
315     <xs:maxLength value="1024" />
316   </xs:restriction>
317 </xs:simpleType>
318 <xs:complexType name="NdicFcdExtension">
319   <xs:sequence>
320     <xs:element name="vehicleType" type="D2LogicalModel:VehicleTypeEnum" minOccurs=
321 ↪ "0" maxOccurs="3" />
322     <xs:element name="trafficLevel" type="D2LogicalModel:TrafficLevel" minOccurs="0
323 ↪ " />
324     <xs:element name="queueInformation" type="D2LogicalModel:QueueInformation"

```

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320     </xs:sequence>
321 </xs:complexType>
322 <xs:simpleType name="NonNegativeInteger">
323   <xs:restriction base="xs:nonNegativeInteger" />
324 </xs:simpleType>
325 <xs:complexType name="NumberOfFcdVehicles">
326   <xs:sequence>
327     <xs:element name="vehicleGroup" type="D2LogicalModel:VehicleGroup" maxOccurs=
328     ↪ "unbounded" />
329     <xs:element name="numberOfFcdVehiclesExtension" type="D2LogicalModel:_
330     ↪ ExtensionType" minOccurs="0" />
331   </xs:sequence>
332 </xs:complexType>
333 <xs:complexType name="PayloadPublication" abstract="true">
334   <xs:sequence>
335     <xs:element name="feedType" type="D2LogicalModel:String" minOccurs="0" ↪
336     ↪ maxOccurs="1" />
337     <xs:element name="publicationTime" type="D2LogicalModel:DateTime" minOccurs="1" ↪
338     ↪ maxOccurs="1" />
339     <xs:element name="publicationCreator" type=
340     ↪ "D2LogicalModel:InternationalIdentifier" />
341     <xs:element name="payloadPublicationExtension" type="D2LogicalModel:_
342     ↪ ExtensionType" minOccurs="0" />
343   </xs:sequence>
344   <xs:attribute name="lang" type="D2LogicalModel:Language" use="required" />
345 </xs:complexType>
346 <xs:simpleType name="Percentage">
347   <xs:restriction base="D2LogicalModel:Float" />
348 </xs:simpleType>
349 <xs:complexType name="PercentageValue">
350   <xs:complexContent>
351     <xs:extension base="D2LogicalModel:DataValue">
352       <xs:sequence>
353         <xs:element name="percentage" type="D2LogicalModel:Percentage" minOccurs="1"
354         ↪ " maxOccurs="1" />
355         <xs:element name="percentageValueExtension" type="D2LogicalModel:_
356         ↪ ExtensionType" minOccurs="0" />
357       </xs:sequence>
358     </xs:extension>
359   </xs:complexContent>
360 </xs:complexType>
361 <xs:complexType name="QualityParameters">
362   <xs:sequence>
363     <xs:element name="supplierCalculatedDataQuality" type="D2LogicalModel:Percentage
364     ↪ " minOccurs="1" maxOccurs="1" />
365     <xs:element name="numberOfFcdVehicles" type="D2LogicalModel:NumberOfFcdVehicles
366     ↪ " minOccurs="0" />
367   </xs:sequence>
368 </xs:complexType>
369 <xs:complexType name="Queue">
370   <xs:complexContent>
371     <xs:extension base="D2LogicalModel:DataValue">
372       <xs:sequence>
373         <xs:element name="queueLength" type=
374         ↪ "D2LogicalModel:MetresAsNonNegativeInteger" minOccurs="1" maxOccurs="1" />

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364     <xs:element name="fromPoint" type="D2LogicalModel:Percentage" minOccurs="0"
↪ maxOccurs="1" />
365     <xs:element name="toPoint" type="D2LogicalModel:Percentage" minOccurs="0"
↪ maxOccurs="1" />
366     <xs:element name="queueExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
367     </xs:sequence>
368   </xs:extension>
369 </xs:complexContent>
370 </xs:complexType>
371 <xs:complexType name="QueueInformation">
372   <xs:sequence>
373     <xs:element name="queue" type="D2LogicalModel:Queue" maxOccurs="unbounded" />
374     <xs:element name="queueInformationExtension" type="D2LogicalModel:_ExtensionType"
↪ " minOccurs="0" />
375   </xs:sequence>
376 </xs:complexType>
377 <xs:simpleType name="Seconds">
378   <xs:restriction base="D2LogicalModel:Float" />
379 </xs:simpleType>
380 <xs:complexType name="Source">
381   <xs:sequence>
382     <xs:element name="sourceIdentification" type="D2LogicalModel:String" minOccurs=
↪ "0" maxOccurs="1" />
383     <xs:element name="sourceExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
384   </xs:sequence>
385 </xs:complexType>
386 <xs:complexType name="SpeedPercentile">
387   <xs:sequence>
388     <xs:element name="vehiclePercentage" type="D2LogicalModel:PercentageValue" />
389     <xs:element name="speedPercentile" type="D2LogicalModel:SpeedValue" />
390     <xs:element name="speedPercentileExtension" type="D2LogicalModel:_ExtensionType"
↪ " minOccurs="0" />
391   </xs:sequence>
392 </xs:complexType>
393 <xs:complexType name="SpeedValue">
394   <xs:complexContent>
395     <xs:extension base="D2LogicalModel:DataValue">
396       <xs:sequence>
397         <xs:element name="speed" type="D2LogicalModel:KilometresPerHour" minOccurs=
↪ "1" maxOccurs="1" />
398         <xs:element name="speedValueExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
399       </xs:sequence>
400     </xs:extension>
401   </xs:complexContent>
402 </xs:complexType>
403 <xs:simpleType name="String">
404   <xs:restriction base="xs:string">
405     <xs:maxLength value="1024" />
406   </xs:restriction>
407 </xs:simpleType>
408 <xs:complexType name="TrafficData" abstract="true">
409   <xs:complexContent>
410     <xs:extension base="D2LogicalModel:BasicData">

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411     <xs:sequence>
412       <xs:element name="forVehiclesWithCharacteristicsOf" type=
↪ "D2LogicalModel:VehicleCharacteristics" minOccurs="0" />
413       <xs:element name="trafficDataExtension" type="D2LogicalModel:_ExtensionType"
↪ " minOccurs="0" />
414     </xs:sequence>
415   </xs:extension>
416 </xs:complexContent>
417 </xs:complexType>
418 <xs:complexType name="TrafficLevel">
419   <xs:complexContent>
420     <xs:extension base="D2LogicalModel:DataValue">
421       <xs:sequence>
422         <xs:element name="trafficLevelValue" type="D2LogicalModel:TrafficLevelEnum"
↪ minOccurs="1" maxOccurs="1" />
423         <xs:element name="trafficLevelExtension" type="D2LogicalModel:_ExtensionType"
↪ " minOccurs="0" />
424       </xs:sequence>
425     </xs:extension>
426 </xs:complexContent>
427 </xs:complexType>
428 <xs:simpleType name="TrafficLevelEnum">
429   <xs:restriction base="xs:string" />
430 </xs:simpleType>
431 <xs:complexType name="TrafficSpeed">
432   <xs:complexContent>
433     <xs:extension base="D2LogicalModel:TrafficData">
434       <xs:sequence>
435         <xs:element name="averageVehicleSpeed" type="D2LogicalModel:SpeedValue"
↪ minOccurs="0" />
436         <xs:element name="speedPercentile" type="D2LogicalModel:SpeedPercentile"
↪ minOccurs="0" />
437         <xs:element name="trafficSpeedExtension" type="D2LogicalModel:_ExtensionType"
↪ " minOccurs="0" />
438       </xs:sequence>
439     </xs:extension>
440 </xs:complexContent>
441 </xs:complexType>
442 <xs:complexType name="TrafficStatus">
443   <xs:complexContent>
444     <xs:extension base="D2LogicalModel:BasicData">
445       <xs:sequence>
446         <xs:element name="trafficStatus" type="D2LogicalModel:TrafficStatusValue"
↪ minOccurs="0" />
447         <xs:element name="trafficStatusExtension" type="D2LogicalModel:_
↪ TrafficStatusExtensionType" minOccurs="0" />
448       </xs:sequence>
449     </xs:extension>
450 </xs:complexContent>
451 </xs:complexType>
452 <xs:simpleType name="TrafficStatusEnum">
453   <xs:restriction base="xs:string">
454     <xs:enumeration value="impossible" />
455     <xs:enumeration value="congested" />
456     <xs:enumeration value="heavy" />
457     <xs:enumeration value="freeFlow" />

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458     <xs:enumeration value="unknown" />
459   </xs:restriction>
460 </xs:simpleType>
461 <xs:complexType name="TrafficStatusValue">
462   <xs:complexContent>
463     <xs:extension base="D2LogicalModel:DataValue">
464       <xs:sequence>
465         <xs:element name="trafficStatusValue" type="D2LogicalModel:TrafficStatusEnum
466 ↪ " minOccurs="1" maxOccurs="1" />
467         <xs:element name="trafficStatusValueExtension" type="D2LogicalModel:_
468 ↪ ExtensionType" minOccurs="0" />
469       </xs:sequence>
470     </xs:extension>
471   </xs:complexContent>
472 </xs:complexType>
473 <xs:complexType name="TravelTimeData">
474   <xs:complexContent>
475     <xs:extension base="D2LogicalModel:BasicData">
476       <xs:sequence>
477         <xs:element name="travelTimeType" type="D2LogicalModel:TravelTimeTypeEnum"
478 ↪ minOccurs="0" maxOccurs="1" />
479         <xs:element name="vehicleType" type="D2LogicalModel:VehicleTypeEnum"
480 ↪ minOccurs="0" maxOccurs="unbounded" />
481         <xs:element name="travelTime" type="D2LogicalModel:DurationValue" minOccurs=
482 ↪ "0" />
483         <xs:element name="freeFlowTravelTime" type="D2LogicalModel:DurationValue"
484 ↪ minOccurs="0" />
485         <xs:element name="normallyExpectedTravelTime" type=
486 ↪ "D2LogicalModel:DurationValue" minOccurs="0" />
487         <xs:element name="freeFlowSpeed" type="D2LogicalModel:SpeedValue" minOccurs=
488 ↪ "0" />
489         <xs:element name="travelTimeDataExtension" type="D2LogicalModel:_
490 ↪ TravelTimeDataExtensionType" minOccurs="0" />
491       </xs:sequence>
492     </xs:extension>
493   </xs:complexContent>
494 </xs:complexType>
495 <xs:simpleType name="TravelTimeTypeEnum">
496   <xs:restriction base="xs:string">
497     <xs:enumeration value="best" />
498     <xs:enumeration value="estimated" />
499     <xs:enumeration value="instantaneous" />
500     <xs:enumeration value="reconstituted" />
501   </xs:restriction>
502 </xs:simpleType>
503 <xs:complexType name="VehicleCharacteristics">
504   <xs:sequence>
505     <xs:element name="vehicleType" type="D2LogicalModel:VehicleTypeEnum" minOccurs=
506 ↪ "0" maxOccurs="unbounded" />
507     <xs:element name="vehicleCharacteristicsExtension" type="D2LogicalModel:_
508 ↪ ExtensionType" minOccurs="0" />
509   </xs:sequence>
510 </xs:complexType>
511 <xs:complexType name="VehicleGroup">
512   <xs:sequence>
513     <xs:element name="count" type="D2LogicalModel:NonNegativeInteger" minOccurs="1"
514 ↪

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503   <xs:element name="vehicleType" type="D2LogicalModel:VehicleTypeEnum" minOccurs=
504   <xs:element name="vehicleGroupExtension" type="D2LogicalModel:_ExtensionType"
505   </xs:sequence>
506 </xs:complexType>
507 <xs:simpleType name="VehicleTypeEnum">
508   <xs:restriction base="xs:string">
509     <xs:enumeration value="anyVehicle" />
510     <xs:enumeration value="car" />
511     <xs:enumeration value="lorry" />
512   </xs:restriction>
513 </xs:simpleType>
514 <xs:complexType name="VersionedReference">
515   <xs:attribute name="id" type="xs:string" use="required" />
516   <xs:attribute name="version" type="xs:string" use="required" />
517 </xs:complexType>
518 <xs:complexType name="VmsFault">
519   <xs:complexContent>
520     <xs:extension base="D2LogicalModel:Fault">
521       <xs:sequence>
522         <xs:element name="vmsFault" type="D2LogicalModel:VmsFaultEnum" minOccurs="1
523         <xs:element name="vmsFaultExtension" type="D2LogicalModel:_ExtensionType"
524         </xs:sequence>
525       </xs:extension>
526     </xs:complexContent>
527 </xs:complexType>
528 <xs:simpleType name="VmsFaultEnum">
529   <xs:restriction base="xs:string">
530     <xs:enumeration value="communicationsFailure" />
531     <xs:enumeration value="incorrectMessageDisplayed" />
532     <xs:enumeration value="incorrectPictogramDisplayed" />
533     <xs:enumeration value="outOfService" />
534     <xs:enumeration value="powerFailure" />
535     <xs:enumeration value="unableToClearDown" />
536     <xs:enumeration value="unknown" />
537     <xs:enumeration value="other" />
538   </xs:restriction>
539 </xs:simpleType>
540 <xs:complexType name="VmsUnitFault">
541   <xs:complexContent>
542     <xs:extension base="D2LogicalModel:Fault">
543       <xs:sequence>
544         <xs:element name="vmsUnitFault" type="D2LogicalModel:VmsFaultEnum"
545         <xs:element name="vmsUnitFaultExtension" type="D2LogicalModel:_ExtensionType
546         </xs:sequence>
547       </xs:extension>
548     </xs:complexContent>
549 </xs:complexType>
550 </xs:schema>

```

CHANGELOG

version 1.0.0

- Initial format and documentation

BIBLIOGRAPHY

- [16157-1] CEN TS 16157-1:2011 Intelligent transport systems – DATEX II data exchange specifications for traffic management and information – Part 1: Context and framework, https://standards.cencenelec.eu/dyn/www/f?p=CEN:110:0:::FSP_PROJECT:31061&cs=1ED8DA728BFDB3CF4C98EF696427A420D
- [16157-2] CEN TS 16157-2:2011 Intelligent transport systems - DATEX II data exchange specifications for traffic management and information - Part 2: Location referencing, , https://standards.cencenelec.eu/dyn/www/f?p=CEN:110:0:::FSP_PROJECT:31062&cs=11C587676AF23E874354108C5B94B99F6
- [16157-3] CEN TS 16157-3:2011 Intelligent transport systems - DATEX II data exchange specifications for traffic management and information - Part 3: Situation Publication, https://standards.cencenelec.eu/dyn/www/f?p=CEN:110:0:::FSP_PROJECT:31063&cs=154597D857428FFFF988477C51088FE30
- [16157-4] CEN TS 16157-4:2014 Intelligent transport systems - DATEX II data exchange specifications for traffic management and information - Part 4: Variable Message Sign (VMS) Publications, https://standards.cencenelec.eu/dyn/www/f?p=CEN:110:0:::FSP_PROJECT:37597&cs=1046FF87225820240FCDAE9E4EF3118EB
- [16157-5] CEN TS 16157-5:2014 Intelligent transport systems - DATEX II data exchange specifications for traffic management and information - Part 5: Measured and elaborated data publications, https://standards.cencenelec.eu/dyn/www/f?p=CEN:110:0:::FSP_PROJECT:37966&cs=110BD77D7F2B9A32FB04403D101009B89
- [16157-6] CEN TS 16157-6:2015 Intelligent transport systems - DATEX II data exchange specifications for traffic management and information - Part 6: Parking Publications, https://standards.cencenelec.eu/dyn/www/f?p=CEN:110:0:::FSP_PROJECT:40220&cs=1084702C54C56E3547969771C8305250B
- [d2ref] DATEX II Current version/Reference set, <https://docs.datex2.eu/downloads/>.
- [d2supp] DATEX II Current version/Supporting, <https://docs.datex2.eu/>.

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